Prague University of Economics and Business

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BEHAVIORAL TACTICS TO DRIVE SPENDING AND PROLONG PLAY IN THE VIDEO GAME INDUSTRY

BACHELOR THESIS

Study programme: Applied Informatics

Specialization: Applied Informatics

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Prague, May 2025

Acknowledgement

I would like to extend my sincere thanks to Mgr. Ing. Tomáš Sigmund, Ph.D., for supervising my thesis. His insight and comments were very valuable in the creation of this paper. Additionally, I want to thank my family and friends for their unwavering encouragement and support throughout this process.

Abstract

The video game industry’s monetization strategies have transformed from one-time purchases to dynamic “live service” models that leverage behavioral psychology and predatory techniques to drive ongoing player engagement and recurring spending. The main goal of this bachelor thesis is to consolidate research about these psychological methods and how they are used in the video game industry. The first part reviews key psychological techniques and describes how they are embedded in specific monetization models. The second part empirically analyzes player experience, gaming and spending habits, and awareness of these psychological methods through quantitative research. By consolidating fragmented literature and combining it with original empirical data, this paper provides a structured overview of how behavioral psychology is commodified in gaming. The findings offer actionable insights for players, game developers and regulators aiming to understand and ethically govern the psychological design tactics found in video games.

Keywords

flow theory, video games, psychology, monetization

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Introduction

From static one-time payments to dynamic models that emphasize ongoing participation and recurring spending, the video game industry's monetization types have undergone a significant transformation. Present strategies that maximize player engagement and income creation use behavioral psychology and predatory monetization models such as loot boxes and freemium models (Petrovskaya & Zendle, 2022). Principles like the flow theory, which describes a state of total enjoyment and immersion gained when a challenge precisely matches the player's skill level, serve as the foundation for these systems (Csikszentmihalyi, 2008).

These monetization tactics increasingly add variable reward schedules and operant conditioning to drive compulsive behavior in players (Petrovskaya & Zendle, 2022; Vu, 2017). For example, “battle passes” often lock exclusive content behind seasonal/monthly deadlines, incentivizing urgency. These strategies bring up moral questions regarding the use of psychological weaknesses, especially among children and teenagers.

The main goal of this thesis is to systematically identify behavioral design tactics employed by the video game industry to prolong player engagement and drive in-game spending. The first part covers already mentioned psychological techniques like flow theory, FOMO, and “the Skinner box” etc., the description of commonly used predatory monetization systems and how psychological methods are implemented in video games. Data and information regarding all the aforementioned topics are based on relevant research papers, literature, articles, journals, media and the author’s knowledge.

The study’s second part analyzes player perceptions and experiences with psychological tactics via a structured questionnaire combining questions about their gaming habits and Likert-scales to measure vulnerability to compulsion cycles, spending habits and the player’s awareness.

There has been much research done and articles written on these topics separately, however a key contribution of this paper lies in consolidating this fragmented research and providing a structured overview that summarizes past findings while providing empirical insights into player experiences, offering players, developers, and regulators a clearer understanding of how behavioral psychology is commodified in gaming.

1 Flow Theory and Its Application in Video Games

While different psychological tactics (such as FOMO and Skinner box) manipulate behavior through extrinsic rewards or social forces, flow theory explains why players willingly immerse themselves in the game for a period of time, tapping into intrinsic motivation, creating a self-sufficient cycle of difficulty and command. By focusing on flow, this thesis demonstrates how predatory monetization systems of the marketing companies depend on the control of intrinsic motivation. Without the deep immersion flow provides, extrinsic tactics like loot boxes or limited-time offers would lack their coercive power. This dynamic reveals a core ethical dilemma in current game design; while flow is not inherently unethical, intentionally disrupting it for financial gain converts a psychological asset into an exploitable weakness.

1.1 Introduction to Flow

The idea of flow, a fundamental part of modern game design, was originally introduced by a psychologist named Mihaly Csikszentmihalyi. His theory, developed in the 1970s, describes an optimal psychological state where individuals experience intense focus, temporary time distortion and intrinsic motivation during activities that balance their perceived skills with situational challenges (Khoshnoud et al., 2020). Because video games are inherently interactive and goal-oriented, flow offers situations for developing experiences that engage players for extended periods of time.

Csikszentmihalyi (2008) identified eight core components of enjoyment:

- **Achievable tasks:** Confronting tasks have to be within a person's capacity to complete.
- **Focused concentration:** The challenge requires one's intense focus on the activity, blocking out other distractions.
- **Clear goals:** The task at hand must have defined objectives that guide the person's actions.
- **Direct and immediate feedback:** Progress is communicated to people in real time, reinforcing their engagement.
- **Effortless involvement:** Complete absorption in the activity, dissolving mundane preoccupations via heightened attentional focus.
- **Sense of control:** People feel agency over their actions, even in challenging situations.
- **Loss of self-consciousness:** Temporary disappearance of self-awareness during the activity, followed by a stronger sense of self afterwards.
- **Altered time perception:** Time feels distorted, minutes feel like hours or hours pass like minutes.

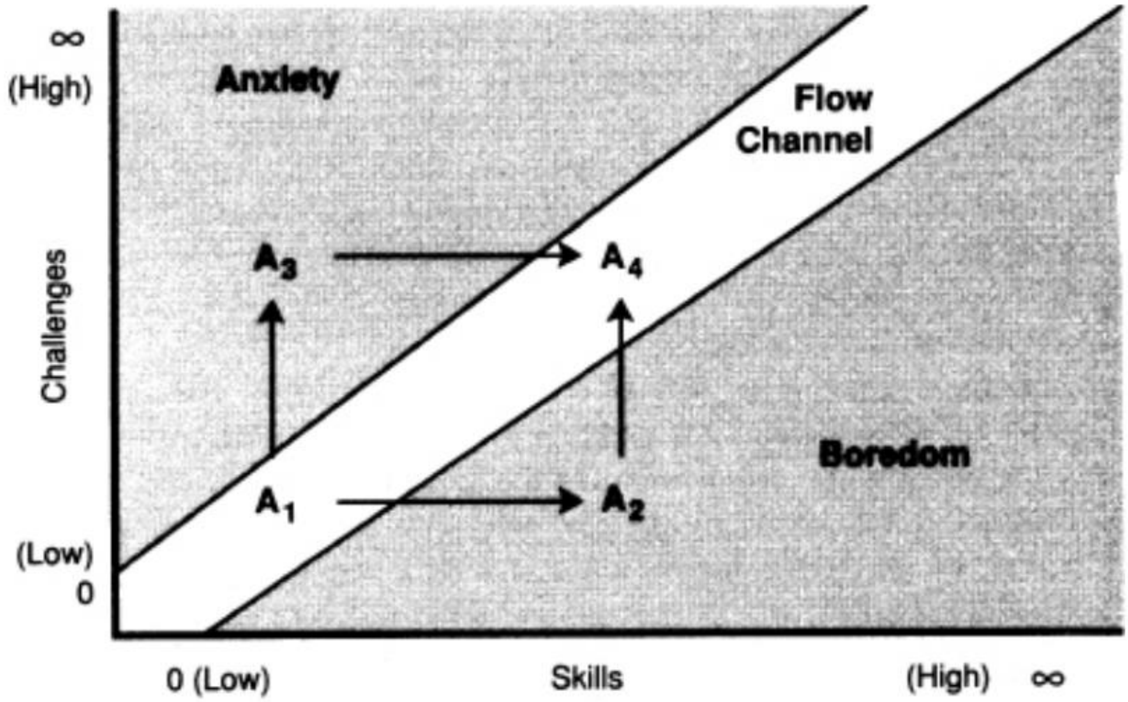
In video games, these components appear via in-game goals (e.g., completing levels), real-time feedback (e.g., score updates) and progressively increasing challenge difficulties that correspond with the growth of players' skill (Khoshnoud et al., 2020; Larche & Dixon, 2020). Neurocognitive studies reveal that flow states correlate with increased activation in the prefrontal cortex, which controls executive functions such as decision-making, and decreased activity in the default mode network, associated with self-referential processing (Khoshnoud et al., 2020). This neural reconfiguration furthers the "loss of self" described in flow, allowing players to fully immerse themselves in the virtual environment.

Huang (2023) mentions that if a game has a well thought out flow system, even games with straightforward designs or unimpressive aesthetics can draw a sizable player base. Conveying a meaningful experience to the player is ultimately the main goal of a game, two necessary parts of a successful game. Without investing a lot of money in creating an emotional story or incredibly lifelike visuals that demand players' full attention, developers can effectively satisfy the entertainment component by using a flow pattern.

1.2 Foundations of the Flow Theory

The "Flow Channel" concept defines the skill-challenge balance, which is key to the use of flow in game design (Larche & Dixon, 2020). According to this framework, the best level of enjoyment happens when a game's difficulty (challenge) corresponds with the player's ability (skill). Csikszentmihalyi uses a chart to illustrate this relationship, with the y-axis showing the degree of difficulty and the x-axis showing the individual's skill. The diagonal zone where skill and challenge stay proportionate and keep people from going into "boredom" or "anxiety" is known as the "Flow Channel".

Figure 1 - *Flow diagram (Source: Csikszentmihalyi, 2008)*

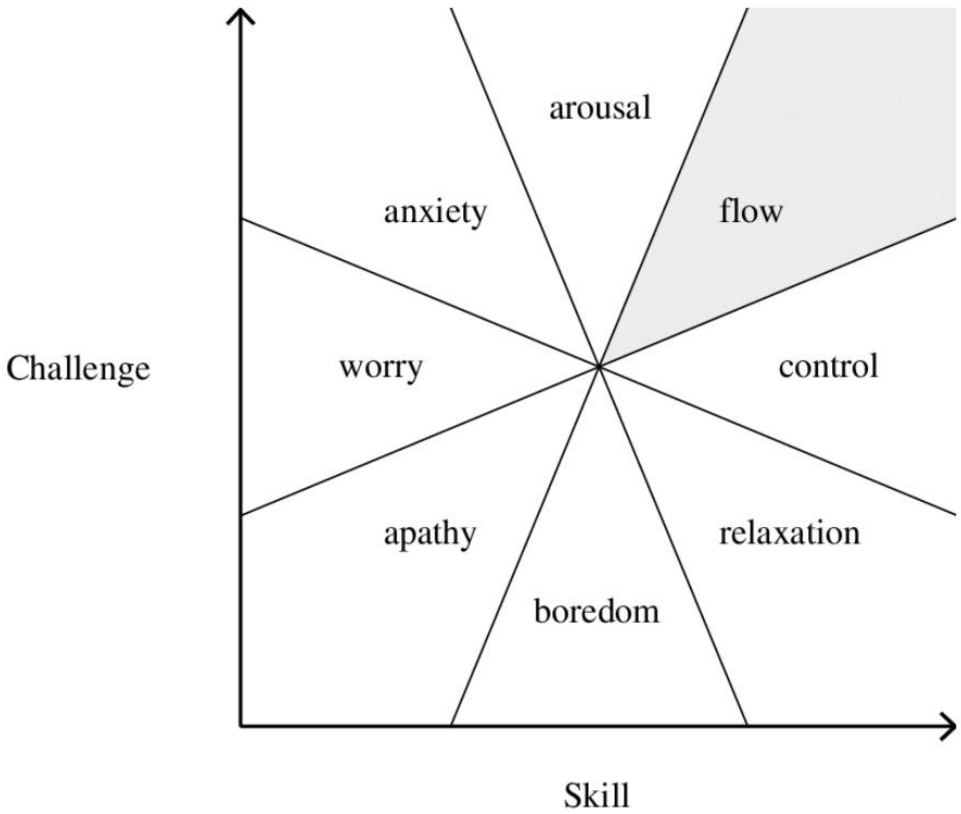


The figure above represents what most people will experience during a task. Point A1 is when the person is still new to the activity so their skill is also low, and the intensity of the challenge should also be low to achieve the flow state. Over time, the person's skills will improve and if the challenge intensity does not increase the person will get bored (A2). On the contrary, if the challenge at hand is too hard and the person's ability is low, they will get anxious (A3). That is why the activity's difficulty should simultaneously change according to the person's growing skills and remain in the "Flow Channel" (A4).

It is important to note that while both points A1 and A4 are in the diagonal zone of the flow, each point means a much different level of experience. Point A4 is a much more intricate experience since the challenge and skill became much more complex (Csikszentmihalyi, 2008).

The flow chart has undergone different changes and variations throughout the years. One of the more prominent diagrams nowadays can be seen in Figure 2 adapted from Csikszentmihalyi's book *Finding Flow: The Psychology of Engagement with Everyday Life* made in 1997. The diagram that was shown in Figure 1 was made in 1990. This new chart shows far more type of emotions that players can feel than the one prior (except for anxiety, boredom and flow). Other intermediate phases were added including anxiety, relaxation, control, and arousal are highlighted in this updated diagram. The modified chart depicts a wider range of the emotional continuum by mapping these differences on the skill-challenge axis. This shows how people can vary their emotional states based on little adjustments in ability or difficulty.

Figure 2 - *Flow diagram adapted from Finding flow: The psychology of engagement with everyday life (1997) (Source: Brady, 2018)*



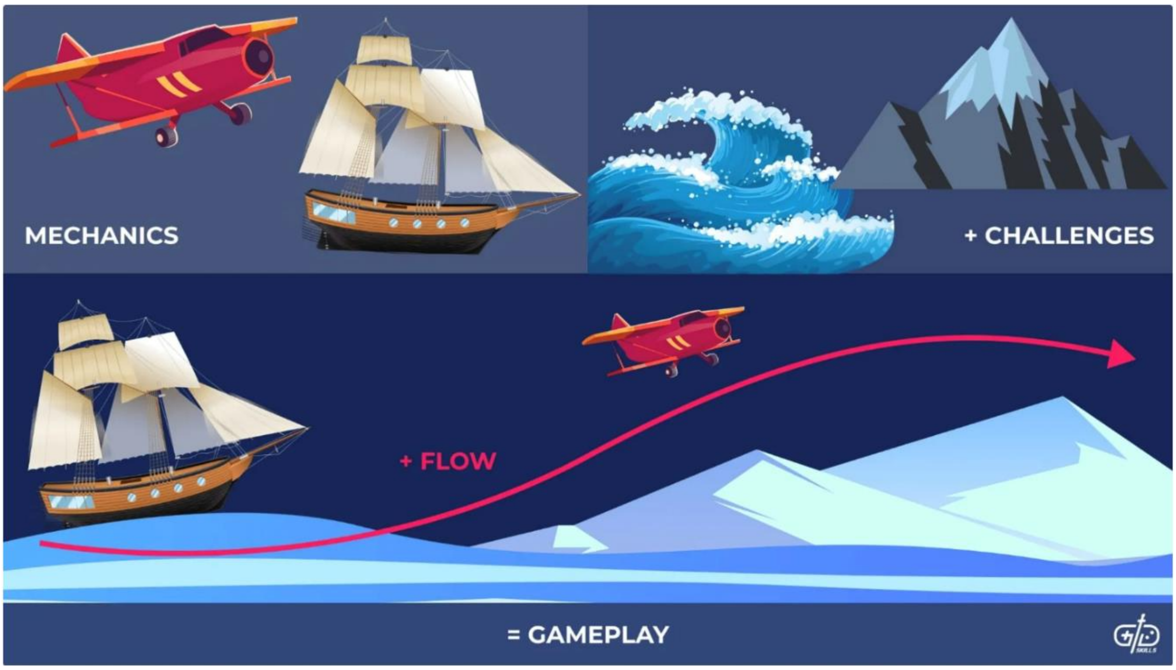
1.3 Flow Components in Video Games

Two primary components significantly influence the degree to which gameplay achieves a state of flow; the design of game mechanics and the use of visual and audio cues (Brazie, 2024).

1.3.1 Game Mechanics

According to Brazie (2024), if the game's mechanics are flawed, for example, if a player's character frequently fails to perform basic movements which disrupts the player's progression, preventing them to achieve flow. Well-designed mechanics are those that use intense movement or combat while remaining accessible to a casual player. Meaning they must be challenging enough to engage the player without becoming excessively difficult. "Gameplay flow" can be understood as the seamless transition between challenges, the operational mechanics and the following outcomes as can be seen in Figure 3. This is similar to the smooth narrative progression in a compelling film, where there is no downtime between successive actions (Brazie, 2024).

Figure 3 - *Mechanics and challenges making gameplay flow (Source: Brazie, 2024)*



1.3.2 Visual and Audio Cues

Visual and audio cues also play a key role in maintaining player's flow. The clearness of these cues might range from tiny hints to obvious messages like distinguishing color markings. Indicators of possible interactive opportunities can be, for example, textured walls with visible vines or contrasting lighting (e.g., showing a climbable surface or warning against a slippery area). Similar to this, changes in the music can warn players of incoming danger or set the mood for action-packed game segments. In the game *The Legend of Zelda: Ocarina of Time*, the player's companion will scream "Hey! Listen!" loudly to notify the player when they missed something (Brazie, 2024).

When combined, these two elements are what encourage players to keep playing the game. Brazie (2024) says, that a well-balanced combination of game mechanics and sensory cues encourages exploration and engagement while reducing the player's overthinking. This dynamic is further reinforced by in-game rewards, which provide players with concrete incentives to keep playing, such as experience points, new tools, or portable items. This kind of drive reflects the innate human desire to explore, which is heightened in the gaming environment due to the lack of real-world dangers (Brazie, 2024).

1.4 Dynamic Difficulty Adjustment (DDA)

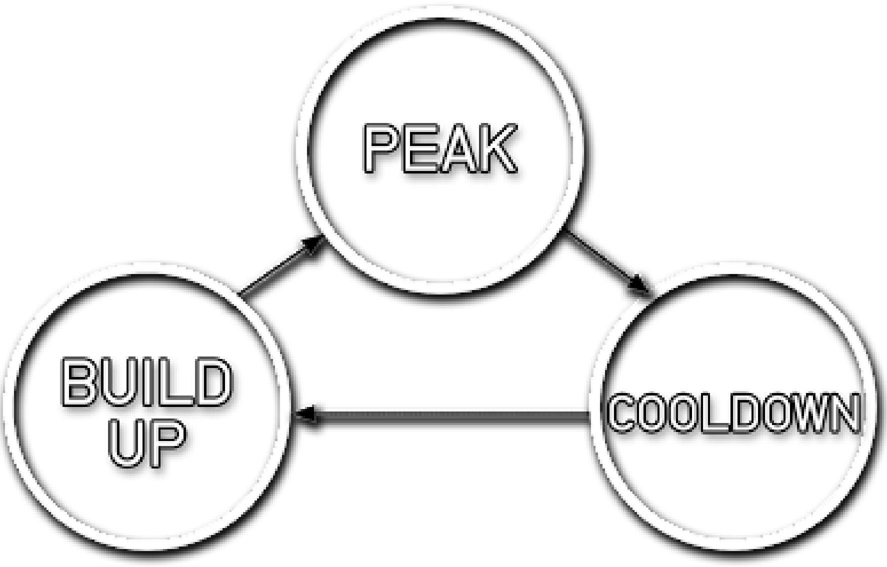
One of the most direct uses of flow theory in video game design is Dynamic Difficulty Adjustment (DDA), which is based on algorithms that analyze player behavior. DDA is defined as *a method of automatically modifying a game's features, behaviors, and scenarios in real-time, depending on the player's skill, so that the player, when the game is very simple, does not feel bored or frustrated, when it is very difficult* (Zohaib, 2018).

Many games typically increase difficulty linearly or gradually during gameplay with features such as starting levels or rates set only at the beginning by choosing a preselected difficulty level. This approach can create a negative experience for players since they can choose a difficulty that may not match their individual skill level and development (Zohaib, 2018). DDA solves this problem by presenting a solution that maintains the balance between challenge and skill, the core requirement for achieving the flow state.

1.4.1 Games that use DDA

Left 4 Dead is an example of a game that successfully implemented DDA within their gameplay loop using their artificial intelligence called "The Director". In this game players are fighting and trying to survive through a zombie apocalypse using guns and other types of weapons. The director is responsible for regulating the intensity of the game by increasing the number of enemies the players have to fight against (Thompson, 2022).

Figure 4 - *States of the director in Left 4 Dead (Source: Thompson, 2022)*



The director can either spawn more zombies to go after the players randomly or choose to target a specific player. The system goes through three states as shown in Figure 4. The build-up phase is where it increases the players' anxiety based on their current level of intensity. The peak is where the director will bring the stress level of the players to the maximum by adding as many zombies as possible. Finally, in the cooldown state it waits before spawning more zombies to give players time to restock weapons and relax, bringing their anxiety levels down to normal (Thompson, 2022).

Figure 5 - *Screenshot of a boulder chasing the player in Crash Bandicoot 2 (Source: Author)*

Another classic example of a game that uses DDA is the renowned platformer game called Crash Bandicoot 2: Cortex Strikes. The developer of this game, Naughty Dog, added this system because in their first game many players were unhappy with the most difficult levels of the game being unenjoyable and anxiety inducing for less skilled players (Baranowski, 2018). In a 2011 blog article the co-founder mentioned, "*Our mantra became help weaker players without changing the game for the better players. We called all this DDA, Dynamic Difficulty Adjustment, and at the time the extent to which we did it was pretty novel. It would lead later Crash games to be the inclusive, perfectly balanced games they became. Good player, bad player, everyone loved Crash games. They never realized it is because they were all playing a slightly different game, balanced for their specific needs*" (Gavin, 2011).

The game would then track how many times the player failed the challenge and adjust the difficulty accordingly. For example, if a player died too many times running from a boulder, as seen in Figure 5, the game would slow down the boulder each time a player failed. Gradually implementing more features, like adding more checkpoints, in-game items giving invincibility or other power-ups, would ease the game for worse players while simultaneously not affecting the more skilled ones (Baranowski, 2018).

2 Psychological Methods Used to Prolong Play and Drive Spending

The goal of flow theory is to balance player ability and challenges in order to keep players' interest. However, additional strategies frequently complement this balance, either by enhancing the experience or by using behavior tactics to encourage players to spend money and time. There might be a risk of undermining flow with using these methods if they induce anxiety, distraction, or guilt. Game design must weave these methods around, not in place of, flow's core ingredients, ensuring players remain motivated by enjoyment and mastery rather than solely by external pressures. This section covers the way in which game producers employ psychological techniques to prolong player engagement and promote spending. Some of these tactics have been heavily criticized because they amplify a condition, even recognized by the WHO, called the Gaming disorder (Entheosweb, 2025). In the ICD-11[1], gaming disorder is characterized by a pattern of digital or video gaming behavior in which individuals struggle to control their gaming habits, increasingly favor gaming over other important activities and interests, and continue or even intensify their gaming despite experiencing negative consequences (World Health Organization, n.d.).

2.1 Fear of Missing Out (FOMO)

Major video game designers often use the psychological phenomenon called "Fear of Missing Out" (FOMO) which is based on people's fear of missing an amazing experience or events that others are participating in, and they might be absent from, therefore being scared of feeling regret later on. Developers design systems that encourage obsessive participation, and therefore retention, by taking advantage of people's unconscious bias. Entheosweb (2025) mentions that according to Forbes, even a 5% increase in customer retention also increases profits by 25%-95%.

2.1.1 How FOMO is used in Video Games

The use of FOMO in gaming is effective because it taps into basic human motivations and drives. In a research conducted by Przybylski et al. (2013), drawing from the Self-Determination Theory[2], it is stated that FOMO occurs when individuals feel a lack of control over their decisions, ability to achieve goals or relationships with others. For example, the

[1] 11th Revision of the International Classification of Diseases

[2] Self-determination theory posits that optimal human functioning arises when the innate needs for autonomy, competence, and relatedness are fulfilled, fostering intrinsic motivation and overall well-being. Source: Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist, 55*(1), 68–78. https://doi.org/10.1037/0003-066X.55.1.68

gaming industry exploits this tactic by limiting access to content unless players adhere to strict developer schedules with time-limited events and exclusive rewards. Also, when players fall behind their peers, features like guild systems and in-game leaderboards may cause them to feel inferior or excluded. This creates a feedback loop where FOMO drives engagement to alleviate psychological discomfort rather than enhance enjoyment (Przybylski et al., 2013). As mentioned by Andersen & MoldStud Research Team (2024), players often have a tendency to, or a preference, to avoid losses rather than acquiring gains of equivalent value. This cognitive bias is called “loss aversion” and it can be used by game developers to create situations for players that feel urgent and rewarding because when players see an event or an in-game item that is up for a shorter period of time, their brain gives it higher value (Entheosweb, 2025).

2.2 Operant Conditioning and Skinner Box Design

One of behavioral psychology's most common applications in modern game design is the usage of Skinner Box principles in video games. By using reinforcement schedules, these strategies, which have their roots in B.F. Skinner's work on operant conditioning, create compulsive engagement loops that keep players engrossed, frequently past the point of intrinsic enjoyment. These strategies have been improved by game designers to maintain player engagement and encourage spending by rewarding particular behaviors and building anticipation with changeable incentives. Operant conditioning methods are not intrinsically bad, but when applied incorrectly or excessively, they might lead to detrimental gaming habits, as Daniel Vu (2017) points out in his analysis.

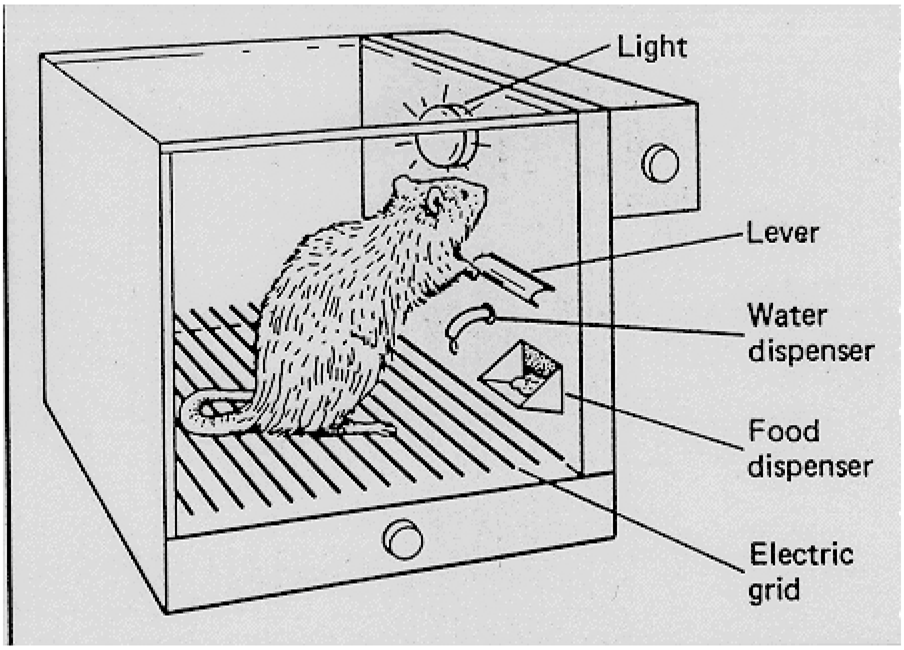
2.2.1 The Foundation of the Skinner Box

Prominent behaviorist psychologist B.F. Skinner developed the idea of operant conditioning, which holds that actions are influenced by their results (Personality at Work, n.d.). He illustrated this in the mid-1900s with the "Skinner Box" (also known as an "operant conditioning chamber"), a soundproof cage, for example rats and other animals. The box featured a lever that, as seen in Figure 6, dispensed food when it was pressed. Skinner noticed that once the action was linked to the incentive of food, the animals learned to press the lever more frequently. Behaviors that result in positive reinforcement (rewards) are likely to be repeated, as seen by this example. Learning is driven by consequences, as demonstrated by the action, reward, and repeated action loop.

In some cases, as described by McLeod (2025), the box also had a grid floor consisting of metal bars capable of delivering mild electric currents. When the lever was pressed the small shocks would stop. The shocks were often paired with stimuli (e.g., lights or sounds) to create conditioned responses and the animals preemptively performed actions to prevent shocks after associating a stimulus, like a light, with impending discomfort.

Skinner’s findings extended beyond animal experiments, influencing human domains such as education, workplace motivation, and even video game design where reward and punishment systems shape behavior.

Figure 6 - *Skinner box experiment involving a rat (Source: Practical Psychology, 2020)*



2.2.2 Reinforcement Schedules in Game Design

Reinforcement schedules play a key role in operant conditioning, dictating how rewards are given out in response to player behavior. Video games make smart use of these "schedules" to increase player involvement.

Games in their early phases frequently employ continuous reinforcement, in which each action results in a reward. Described by Vu (2017), *Candy Crush*[3] uses operant conditioning by rewarding players with bright animations and sound effects when they successfully match patterns, as shown in Figure 7. By making connections between successful gameplay and good feedback, these sensory inputs serve as positive reinforcement. Over time, players subconsciously link these cues to achievement, reinforcing motivation and keeping engagement through a cycle of reward anticipation.

[3] Candy Crush is a popular mobile puzzle game where players match colorful candies in rows or columns to clear levels, developed by a Swedish game developer called King.

Figure 7 - *Visual cue for matching candies in the game Candy Crush (Source: Gordon, 2022)*

Games also often use variable-ratio schedules, which may result in more valued but uncertain prizes. As Personality at Work (n.d.) mentions, this strategy is best shown by loot boxes in video games such as *Overwatch* which provide random prizes for purchases or achievements, generating a sense of gambling anticipation that encourages more expenditure. It is also implemented by unpredictable competitive matches and chance-based systems. This schedule explains why players continue grinding for rare item drops with minimal chance of success or play "just one more match" in competitive games despite fatigue. The unpredictability of rewards creates constant anticipation for future game content without significant positive feedback.

Fixed-interval schedules can be seen in the form of seasonal events or rank resets. For example, players are encouraged to grind matches before the season finishes in order to keep their rank standing because competitive games like Marvel Rivals regularly reset player rankings. Engagement cycles based on external time restrictions are produced by this urgency.

These reinforcement schedules are deliberately put in place to keep players interested while also making it harder to obtain the rewards. Early tasks offer immediate satisfaction, which boosts self-esteem and creates favorable brain pathways linked to success. But as playtime lengthens, rewards take longer to obtain and demand more work.

2.3 Sunk Cost Fallacy

The sunk cost fallacy represents a powerful psychological mechanism that keeps players engaged with games long after they've stopped enjoying them. Emotional investment, not rational evaluation, is the cause behind the sunk cost fallacy. As defined by David McRaney and cited by Adair (n.d.), this fallacy occurs when "*we think we make rational decisions* *based on the future values of objects, investments and experiences, when the truth is our decisions are tainted by emotional investments we accumulate, and the more we invest in something the harder it is to move on from it*." This leads players into a lot of decision traps that keep them playing longer than they would otherwise. Gamers who have devoted hundreds or thousands of hours to character development, thorough learning of the game's system or having in-game assets, experience psychological pressure to keep playing to make up for that commitment. Similarly, those who have purchased numerous games in bundles or sales feel compelled to play them all to rationalize their spending, creating backlogs that themselves become sources of anxiety and obligation rather than pleasure.

Adair (n.d.) mentions that sunk cost fallacy also intertwines with loss aversion which, as stated before with regard to FOMO, is about avoiding losses rather than acquiring gains. Gamers often prioritize staying engaged with games out of fear of losing their accumulated progress, skills, or in-game identities, even though they may ultimately desire a more fulfilling life outside of gaming. It becomes particularly effective in games with progression systems that display accumulated achievements, levels, or statistics, creating concrete manifestations of player investment that would be psychologically painful to abandon.

2.3.1 Game Design Elements Exploiting the Sunk Cost Fallacy

Through a number of complementing mechanisms that produce increasing psychological commitment, game developers purposefully create systems that compound the sunk cost effect. With each hour invested, progression systems create a stronger emotional engagement and demand ever-larger time commitments to advance. When progress percentages clearly show how much has been done, collection techniques produce thorough catalogs that are psychologically difficult to stop working on before finishing. Players can customize landscapes and people with customization tools, resulting in distinctive expressions that increase emotional investment above and beyond objective value. These design components work together to create a psychological environment where disengagement becomes more and more like "wasting" already-committed time and resources.

When it comes to free-to-play games that progressively increase player commitment over time, the effectiveness of sunk cost psychology in gaming is especially evident. There aren't many obstacles to initial engagement, but as players spend weeks and months building collections and advancement, the psychological cost of abandoning becomes significantly higher (King & Delfabbro, 2019). Because they cannot stomach the idea of "wasting" their prior investment, many gamers continue to play games they no longer genuinely like.

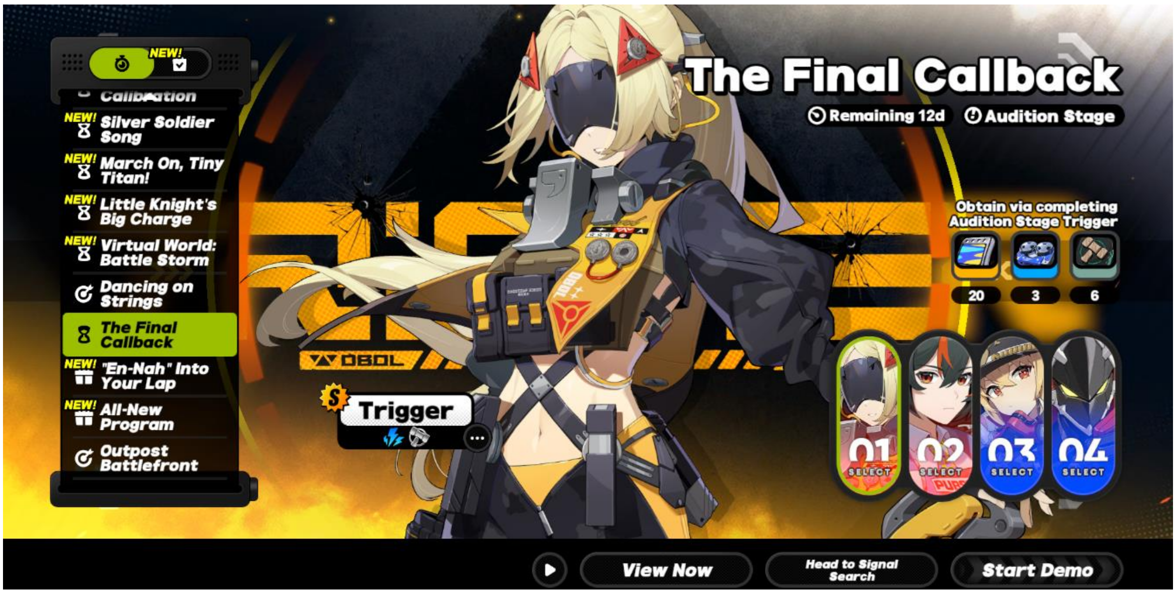
Similarly, players who have paid for a game feel more compelled to continue playing in order to recoup the expense, hence this psychology also holds true for financial investments. By sustaining player involvement through psychological pressures rather than authentic enjoyment and increasing lifetime player value through cognitive biases rather than captivating new experiences, the emotional attachment produced by these sunk cost benefits game creators.

2.4 Endowment effect

The term "endowment effect," which was coined by Nobel laureate Richard Thaler, refers to the human impulse to place an excessive value on things only because we own them (Bassett, 2023). This bias shows itself in product design when consumers form illogical ties to things they make, customize, or temporarily own. This is demonstrated by a seminal study conducted at Duke University, which found that students who won lottery tickets wanted to sell them for 14 times the price that others were prepared to pay to get them (Bassett, 2023). In gaming context, when players obtain virtual in-game assets, their perceived value rises significantly in comparison to the same things they do not own.

For instance, free trial periods for premium features are one way that game designers take advantage of this effect. Games provide players with access to strong items, convenience features, or status changes for a short time, making them unwilling to give them up. In a game called *Zenless Zone Zero*, there are limited-time characters which you can trial, or “demo” as stated in the game, before acquiring them by spending in-game currency, as shown in Figure 8. This makes people have a feeling of ownership of said character and want to have them on their account. Even short-term ownership fosters attachment, which influences purchasing decisions. The effectiveness of "try before you buy" advertising in turning free players into paying customers can be explained by this idea. Loss aversion reactions, that encourage purchases, are triggered when players feel as though they are losing something when they return to basic gameplay after they have gained ownership of a premium character.

Figure 8 - *Character trial (demo) in Zenless Zone Zero (Source: Author)*



As described by Perez (2019), the development team for *Game of Thrones Conquest* tried an event called "Take the Black," which let players trade troops, an important resource in the game, for in-game rewards. They, at the same time, created a troop-training event to encourage participation. Despite these attempts, the feature was canceled after a day because of the strong negative feedback from players. Although the developers were correct to prioritize user feedback, this fiasco demonstrates how ownership biases have a significant impact on player psychology. Players' perception of their troops as irreplaceable assets, valued far above their functional utility, is probably the cause of the intense reaction.

3 Predatory Monetization Systems in Video Games and How They Use Psychological Methods

Although many independent games, such as *Hollow Knight*, or story-driven games like *The Last of Us Part I*, still use conventional, static one-time purchase structures, more and more games are starting to use predatory monetization schemes in live-service, free-to-play and mobile gaming, which this section analyzes. These kinds of systems, like microtransactions, freemium business models and battle passes, are essential components of the game's architecture that put recurring income ahead of player autonomy. This involves purchase structures that obscure the eventual, long-term costs until players have already become both financially and emotionally invested (King & Delfabbro, 2018). This section also covers the involvement of the psychological methods, their use in video games monetization and regulations addressing them, if there are any.

3.1 Freemium Games

Freemium games are free-to-play[4] and earn revenue through a variety of methods, such as in-game ads, microtransactions and subscription-like extras such as battle passes. Many game developers mix these strategies to maximize profit, where players are encouraged to buy enhancements like extra lives, virtual currency, avatar customizations, ad removal or extended playtime (Hubka, 2024). This approach relies on converting a percentage of free players into paying customers.

Freemium games typically create distinct player experiences:

- Non-paying players encounter advertisements, time restrictions, limited customization options, or restricted access to content.
- Paying players receive advantages, conveniences, or exclusive content unavailable to non-paying users.

The model has faced criticism when implementing aggressive "pay-to-win" mechanics, where substantial purchases become virtually necessary to progress or complete the game effectively (Hubka, 2024). Some games create deliberate frustration points that can only be alleviated through payment, effectively using negative experiences as monetization leverage.

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[4] Free-to-play games are those that are available to download and play for free (without an initial purchase).

To give an example of a game using a freemium model, *EverMerge* is a free-to-play merging puzzle game where players combine smaller items into larger ones to expand a magical world filled with fairy tale characters. The core gameplay revolves around merging objects on a limited board while managing resources such as energy and coins that naturally recover over time, by playing the game longer or outright buying them with real money.

“Gems” serve as the paid currency in *EverMerge* and are a key part of its monetization strategy. They are used to refill energy, skip long cooldowns on generators, and accelerate various progression milestones. Because the game intentionally restricts the natural pace of progress, through mechanisms like limited board space and lengthy build timers, gems become a tempting shortcut.

At many decision points, a floating or pop-up prompt may indicate that spending a few gems can push you just past a bottleneck (as can be seen in Figure 9). This design creates a cycle of psychological pressure: the longer you wait, the more progress you risk losing, nudging you to spend gems to avoid frustration.

Figure 9 - *Pop-up in EverMerge to spend paid currency to fast-forward a build timer (Source: Grguric, 2025)*

3.1.1 Psychology Behind Freemium Models

Games like *EverMerge* build urgency by using time-based limits (such as energy systems and cooldown timers) to intentionally slow down progress. Players must decide whether to pay to avoid delays or wait hours for resources to be replenished. This phenomenon called "time-gating" creates FOMO on short-lived incentives while appealing to loss aversion, which is the fear of losing progress or falling behind.

Players place a higher value on virtual items because of artificial scarcity created by, for example, limited board area, limited in-game currency or unique items (such as avatar customization). Players who devote time or money eventually may fall victim to the sunk cost fallacy.

A lot of freemium games purposefully include frustration points elements (such as sudden spikes in difficulty or endless grinding) that can be fixed by paying for them. Players go through unpleasant experiences (e.g., watching advertisements, waiting times) until payment provides comfort which is similar to a pain-reward cycle, tied to Skinner's box. By positioning in-game currency (e.g., *EverMerge's* gems) as the 'lever' to bypass frustration, developers condition players to associate spending with immediate problem-solving.

3.2 Virtual Currency

Virtual currencies serve as intermediaries between real money and in-game purchases, disguising the actual monetary cost of digital items. Games typically implement dual-currency systems:

- premium currency (purchased with real money),
- common currency (earned through gameplay).

The relationship between virtual and real-world currency exchange rates is frequently unclear, making it difficult for players to accurately assess the actual worth of in-game purchases (Toccafondi et al., 2023). Developers often establish these rates with inflated nominal values compared to their real-world monetary equivalents. Toccafondi et al. (2023) mention that this approach can trigger the "money illusion effect" (or face value effect), a cognitive bias where individuals focus on nominal rather than real value. This significantly influences how consumers perceive products and make spending decisions in such contexts.

3.2.1 Psychology Behind Virtual Currencies

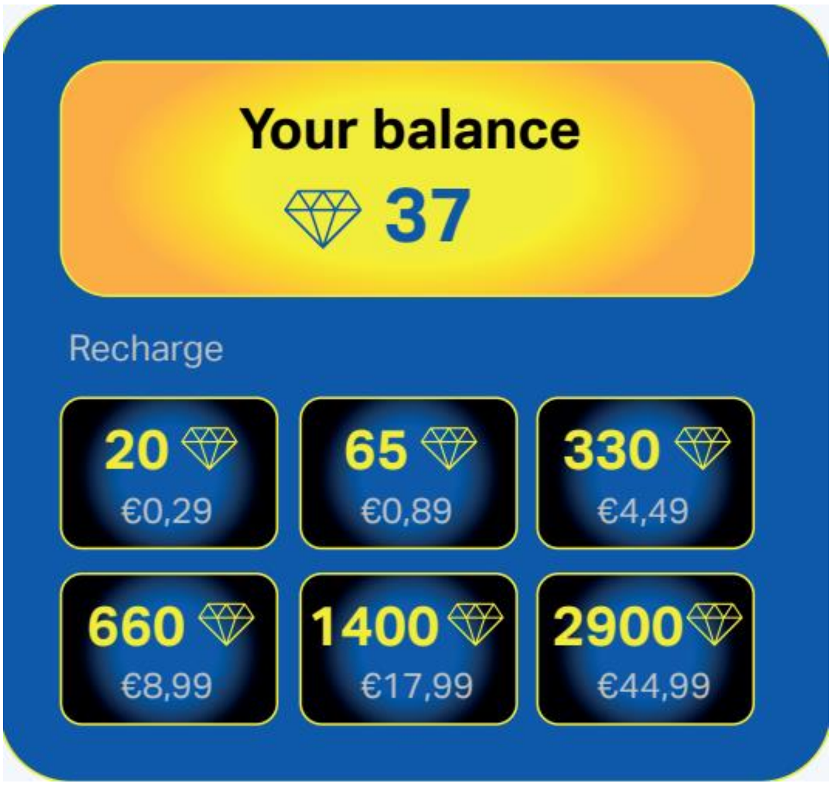
Dual-currency systems encourage users to spend time obtaining common currency which is an example of how they take advantage of the sunk cost fallacy, since this eventually fosters the mindset that "I've put too much effort to quit now," which encourages users to "top-up" their progress by spending premium currency. For instance, a player may argue that, in order to avoid "wasting" their time, they should use real money to purchase the remaining 10% of an item's common currency cost after spending hours earning 90% of it.

Daily in-game rewards in the form of common currency use fixed-interval Skinner box schedules to normalize habitual involvement by teaching participants to log in frequently. Since missing a day "resets" players' progress, streaks (like "7-day bonus") take advantage of loss aversion and put pressure on users to remain consistent. Some players may gradually come to view the daily given currency as limited, which encourages purchases when stockpiles run low.

3.2.2 European Union Guidelines Regarding Virtual Currencies

On March 21, 2025, the Consumer Protection Cooperation (CPC) Network emphasized the need for enhanced consumer safeguards in the European Union's gaming sector regarding in-game virtual currencies. This initiative, aligned with the forthcoming Digital Fairness Act consultation, addresses concerns over exploitative practices affecting vulnerable groups (such as children). For example, some games would offer in-game virtual currencies only in bundles mismatching the value of purchasable in-game digital content and services, as seen in Figure 10, from the document by the Consumer Protection Cooperation Network (2025).

Figure 10 - *Game offering currency only in bundles (Source: Consumer Protection Cooperation Network, 2025)*



The CPC Network outlined non-binding principles to promote transparency and fairness, which are expected to inform national enforcement efforts under existing EU consumer protection laws. These key principles were highlighted by Heinisch (2025):

- **Transparent Pricing**: Games must display real-world monetary values alongside in-game currency costs to empower informed purchasing decisions.
- **Avoiding Deceptive Practices**: Prohibits tactics like multi-currency conversions or obscured pricing structures that confuse players.
- **Consumer Choice**: Players should not be forced to purchase excess virtual currency (for example bundles shown above).
- **Pre-Purchase Clarity**: Traders must provide straightforward information on product details, pricing and withdrawal rights.
- **Withdrawal Rights**: A 14-day withdrawal period applies to unused virtual currency unless waived for immediate access by the player.
- **Fair Contract Terms**: Contracts must use plain language and avoid terms that disproportionately favor developers.
- **Protecting Vulnerable Groups**: Games must integrate safeguards for minors (such as parental controls) and avoid exploiting compulsive spending behaviors.

A Swedish Consumer Association complaint prompted the CPC Network, under the direction of the European Commission, to file a lawsuit against a game developer (Heinisch, 2025). Allegations include aggressive sales methods (such as time-limited offers), deceptive advertising directed at children, and a lack of transparency in pricing. This enforcement demonstrates the EU's dedication to giving minors' protection and consumer rights in gaming top priority. Companies that make video games are advised to conform their operations to EU consumer regulations to reduce the possibility of coordinated investigations and fines (up to 4% of yearly turnover) (Heinisch, 2025).

3.3 Battle Pass

Battle passes operate as a seasonal monetization model where players pay an upfront fee to gain entry to exclusive, time-bound content (Petrovskaya & Zendle, 2022). Players must finish in-game tasks or objectives to unlock the successive tiers of this content. While gameplay involvement is the main factor that determines progression, certain systems allow players to get around these restrictions by making extra payments, which speeds up access to premium rewards.

Battle passes typically feature:

- a free tier with minimal rewards,
- a premium tier with more desirable rewards,
- time-limited availability (usually 1-3 months),
- daily/weekly challenges to encourage regular play.

To give an example, Figure 11 shows how the battle pass looks like in the game called *Fortnite*. The battle pass screen is a horizontal grid of tiers (here, spans 13 pages and 100 levels) split into a free row on top and a premium row below. Player’s current tier and XP-bar sit on the left and it is filled it by playing matches and completing challenges to unlock each column’s rewards. If you purchase the Pass with 950 V-Bucks (*Fortnite’s* premium currency), all locked premium items instantly drop into the player’s Locker (the inventory) and players continue earning new ones as they level up.

Figure 11 - *Battle pass from the game Fortnite (Source: Madigan, 2019)*

3.3.1 Psychology Behind the Battle Pass

Madigan (2019) describes how with its battle pass system, *Fortnite* uses clever psychological tricks to support its free-to-play business model, which brings in billions of dollars annually. The game fundamentally appeals to people's motivations for advancement, dedication, and scarcity. The “principle of progress”, which is backed by studies that demonstrate that quantifiable progress toward objectives increases motivation, is one important motivator (Madigan, 2019). By providing players with continuous, visual feedback, *Fortnite* takes advantage of this. Gaining battle pass benefits like decorative skins or visual effects is made easier with each match, win or loss. In-game notifications are triggered by even minor progress, and players can see how near they are to the next tier.

The “endowed progress effect”, which capitalizes on people’s innate desire to complete the tasks we have begun, is another tactic (Madigan, 2019). *Fortnite* frequently encourages players to make unintentional progress. A notice such as "1 out of 4 chests found" may appear when a player stumbles into a treasure chest, for instance, immediately motivating them to finish the task. "I'm already halfway there, why not keep going?" is what this artificial head start makes the mind believe. By portraying activities as half finished, the game turns leisure play into a mission and instills a sense of duty to complete what has been "started" for them.

The battle pass model creates an "obligation to play after purchases." Players who have invested in a battle pass often feel committed to playing enough to "get their money's worth," even when they might otherwise lose interest in the game. This type of commitment ties back to the sunk cost fallacy described earlier.

Finally, Madigan (2019) mentions how the game uses “artificial scarcity.” The rewards in *Fortnite's* battle pass are subject to temporary "Seasons," usually lasting a few months, after which they disappear forever. By playing on the fear of missing out (FOMO), this generates urgency. Players are under pressure to act fast, either by spending money to skip tiers or by grinding for progress, even if they can claim benefits retrospectively if they purchase the pass after the fact. Additionally, perceived worth is increased by scarcity: demand is fueled by the status symbols that special, limited-edition cosmetics become. The cycle is repeatedly reset by the seasonal rewards, which keeps players interested in the upcoming prizes.

3.4 Loot Box

According to (Webwise, n.d.), loot boxes are virtual items in computer games that hold random, unknown in-game items like character skins, cosmetic upgrades, player icons and more. Gamers can purchase them with real money or by earning them through gameplay achievements. In contrast to typical in-game purchases (such as battle passes or virtual currency), which allow players to choose specific products, loot boxes are based on chance. Players are unaware of the contents until they have obtained and opened them. Because of this element of surprise, the prize could be something that is not as essential or wanted or something really valuable (like uncommon skins).

In *Overwatch 2*, each loot box contains four random cosmetic items like character skins, voice lines, emotes, and weapon charms, categorized by rarity: Common, Rare, Epic, and Legendary. There are regular loot boxes and "legendary" loot boxes which have different drop rates, shown in Figure 12. Every loot box opened guarantees players a Rare item or higher and opening five Loot Boxes consecutively ensures an Epic item, and opening twenty in a row guarantees a Legendary reward (Cale, 2025).

Figure 12 - *Drop rates of loot boxes in Overwatch 2 (Source: Cale, 2025)*

Regular Loot Box Drop Rates	Legendary Loot Box Drop Rates
Common - 97.97%	Common - 97.97%
Rare - 96.26%	Rare - 96.26%
Epic - 21.93%	Epic - 21.93%
Legendary - 5.10%	Legendary - 100%

3.4.1 Psychology Behind the Loot Boxes

Khalid (2023) explores how loot boxes affect the human psyche to keep players engaged and spend money. They are often compared to gambling, arguing they negatively impact younger players and people prone to addictive habits. The article explains how the brain's reaction to unpredictability, fueled by dopamine spikes and FOMO, drives players to chase rare items even when the odds are not in the player's favor.

Subtle strategies are employed by game developers to enhance this impact. They set very low drop rates for coveted products, for example. As a result, gamers continue to purchase additional boxes in the hopes of getting lucky. While virtual currency like "gems" or "coins" conceal how much real money players spend, eye-catching animations and sound effects make opening treasure boxes seem exciting (Khalid, 2023). The sunk cost fallacy is used by bulk purchase agreements, which encourage gamers to ask themselves, "I've already bought three boxes, why not try one more?" Another factor is social pressure, where for example teenagers feel the need to spend money in order to keep up with their peers when they display rare cosmetics.

3.4.2 Regulations Regarding Loot Boxes

In terms of regulating the exploitative nature of the loot boxes, Belgium has taken the strictest stance, where all paid loot boxes are treated as illegal gambling (Xiao, 2024). Publishers must either remove random-reward purchases entirely in Belgium or block Belgian users from accessing those features and from seeing related ads. For example, the game Pokémon Unite decided to completely withdraw their game from the country.

A court ruling in the Netherlands in March 2022 deemed all paid loot boxes, including ones with marketable rewards, to be completely lawful (Xiao, 2024). General consumer protection regulations still apply to loot box sales, meaning you must not employ deceptive "limited-time" deals, comprehensive odds tables must be provided in searchable language, and any storefront (e.g. Google Play Store) or advertisement must explicitly state that randomized purchases are included. The Dutch consumer protection regulator fined the developers of Fortnite over €1.1 million for multiple different violations of the consumer protection law. Across the EU, the Unfair Commercial Practices Directive already enforces these same disclosure rules in Italy, the UK and elsewhere, and the forthcoming Digital Fairness Act is expected to set out even more detailed requirements on loot boxes and in-game currencies (Xiao, 2024).

Players in Austria have both won and lost refund claims over loot boxes with transferable rewards, particularly in the Counter-Strike and EA FC (previously FIFA) series (Xiao, 2024). While some judges affirmed identical contracts as acceptable, others determined that those purchase agreements were unlawful under gambling regulations and so unenforceable, enabling refunds. The Supreme Court is expected to hear an appeal by the summer of 2025, and its decision will most likely influence future events even though it is not legally enforceable. Public coverage of these conflicting outcomes has been distorted by litigation funders who support both publishers and players, emphasizing their individual successes.

Australia and Germany have responded to loot boxes by tightening their age ratings (Xiao, 2024). Every game that offers paid random rewards in Germany is now rated at least a USK 12 (for children aged 12+). While exemptions are theoretically available, none have been granted thus far, and digital stores automatically implement USK 12 by default. This contrasts with PEGI[5] and the ESRB[6], which only record the existence of loot boxes without altering the age group. The same law was implemented in Australia on September 22, 2024. Games that involve simulated gambling features, even without real-money stakes, are categorized as R 18+ and any title that offers loot boxes must have an M rating (suggested for mature audiences - parental discretion advised).

In Mainland China, Taiwan and South Korea, strict probability disclosure rules apply (Xiao, 2024). Developers must display exact drop rates in a format that can be text-searched. Taiwan also requires a specific Chinese disclaimer stating that buying or participating does not guarantee any specific item. Mainland China has already fined developers (for example, CN¥10 000 against Survivor.io) for failing to reveal odds, and South Korean rules even prohibit vague statements like “less than 1%.”

The United Kingdom and Ireland have industry self-regulation and advertising standards requirements (Xiao, 2024). From July 18, 2024, the UK’s Advertising Standards Authority has enforced clear loot box warnings on both app store pages and social media ads. In Ireland, app listings must disclose loot box presence, and while social ads may link to compliant pages, best practice is to include the warning directly.

To protect children online, Spain proposed a new draft law in 2024 that would specifically prohibit anyone under the age of 18 from accessing paid loot boxes with transferable benefits, like those found in games like Counter-Strike and EA FC (Xiao, 2024). Following an unsuccessful attempt to regulate loot boxes in 2022, the new legislation focuses primarily on games that allow in-game items to be traded or sold outside of the game. The majority of gaming companies are unaffected, but those who sell transferable loot boxes have to change to meet the regulations. Brazil is working on a more comprehensive regulation that would use existing criminal rules to declare loot boxes to be unlawful gambling. If approved, the law will forbid loot boxes for both adults and children, marking a stricter, universal ban compared to Spain’s age-restricted approach.

3.5 Gacha Games

The popular mobile game *Puzzle and Dragons* from Japan popularized the gacha game system, a monetization technique utilized in free-to-play games (Game Marketing Genie, 2022). It is named after the "gachapon" toy dispensers and uses in-game currency to acquire randomized in-game products (characters, items, or upgrades). As stated in the article by Game Marketing Genie (2022), players can spend real money to get around time limits but they can also gain currency by simply playing more. Although players can still pay to

[5] PEGI (Pan-European Game Information) is a self-regulatory European video game content rating system, established in April 2003, that uses age categories and content descriptors to give consumers information about the game’s content. (Videogameseurope, n.d.)

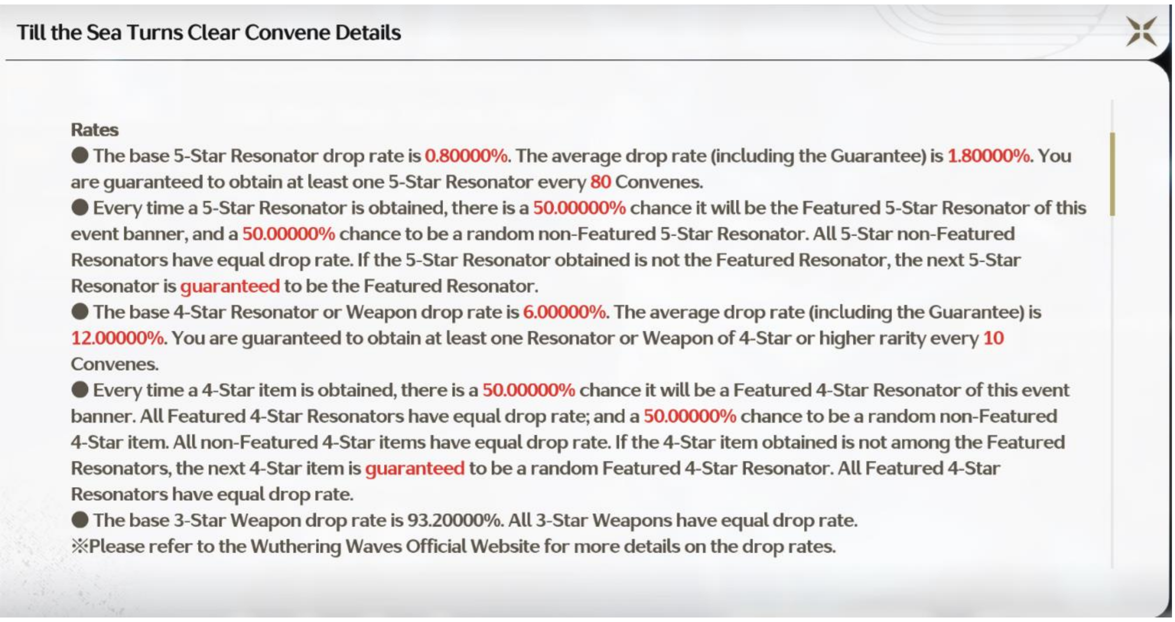
[6] ESRB (Entertainment Software Rating Board) is a North American self-regulatory organization, founded in September 1994, that assigns age-based and content-based ratings to video games so parents and players can determine a title’s suitability. (ESRB, n.d.)

advance more quickly, gachas minimize this type of progression by giving away rewards randomly in contrast to pay-to-win models where purchases ensure advantages.

Gacha games rely heavily on chance, much like spinning a wheel and hoping for a specific outcome. To ease the frustration of repeated bad luck, many incorporate a "pity system" (Gallinule, 2022). This feature makes sure that the game will guarantee the desired reward after a certain number of failed attempts (failing to obtain a rare item or character). Gallinule (2022) also explains that with each try, players may occasionally receive a unique currency that they can later redeem for the item in a special store. Some gacha games employ a counter that in the event that a player makes a certain number of unsuccessful attempts automatically gives them the featured in-game product. By maintaining player engagement while balancing the randomness of gacha mechanics, these systems serve as a safety net.

In the game *Wuthering Waves*, each pull is called a “Convene” and has a tiny 0.8 % base chance to grant a 5-Star Resonator (the characters in the game), which averages out to about 1.8 % once you factor in the pity that guarantees one by 80 pulls, and a 6 % base chance to give a 4-Star Resonator or Weapon (rising to roughly 12 % with a pity every 10 pulls), as shown in Figure 13. Whenever you score a 5-Star or 4-Star, there’s a straight 50/50 shot at it being one of the newly featured Resonators. If you lose that flip, your very next pull of that rarity is guaranteed to be the banner character. Everything else is a 3-Star Weapon, so while buying more currency simply lets you open Convenes faster, both paying players and free-to-play users eventually work their way toward those high-rarity guarantees.

Figure 13 - *Drop rates in the game Wuthering Waves (Source: Author)*

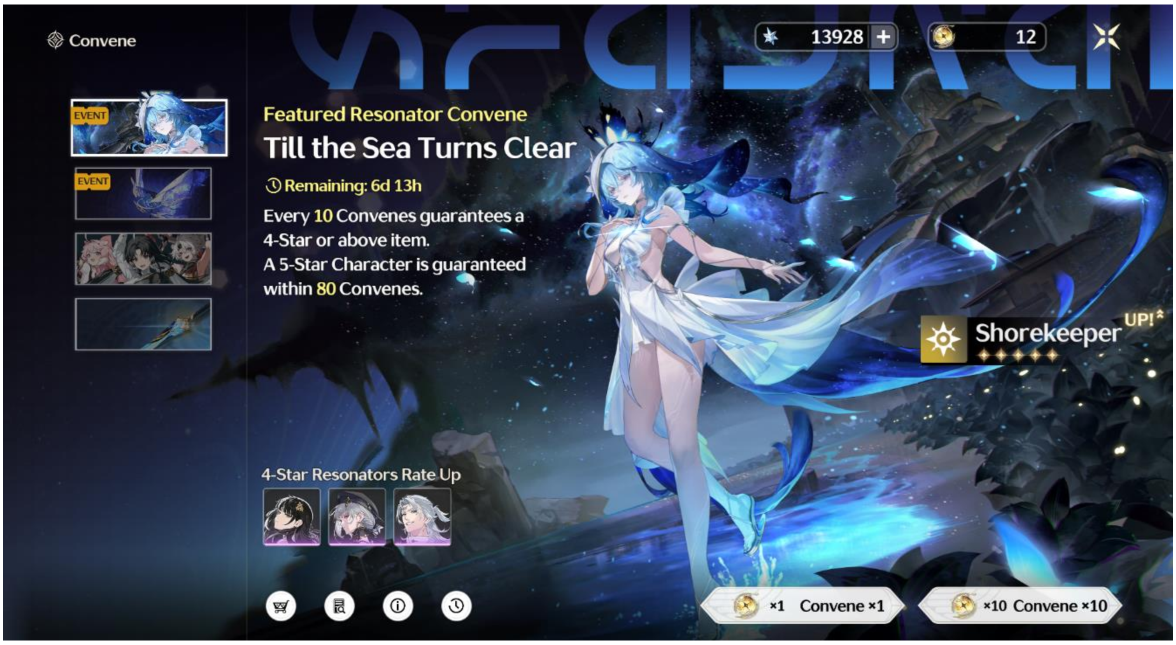


3.5.1 Psychology Behind Gacha Games

In their article, Komad (2023) investigates the psychological basis of gacha mechanics in games today, focusing on the variable-ratio schedule in B.F. Skinner's operant conditioning theory. This reinforcement schedule produces an engagement loop since rewards are given after a specific number of actions. The article highlights how, in spite of the possibility of lengthy unrewarded efforts, the excitement of uncertainty combined with intermittent reinforcement maintains high player retention and spending.

These features in modern gacha games are frequently combined with time-limited events, which incite a sense of urgency and FOMO in players, forcing them to take immediate action or risk losing access to special rewards (Games Learning Society, 2023). This is present for example in *Wuthering* Waves where we can see, in Figure 14, a limited-time character banner's description saying "Remaining: 6d 13h" until it is gone.

Figure 14 - *Character banner in Wuthering Waves (Source: Author)*



Komad (2023) also described how the "near-miss effect" affects players' mental health by generating anxiety when results barely meet expectations, like narrowly missing a rare prize. Players are encouraged to keep trying by this dissonance between expectations and reality which motivates them to spend more time or money in the hopes of eventually succeeding. The sunk cost fallacy is also coupled with this. When combined, these mechanisms drive players to resolve near-successes and prevent "wasting" previous efforts, trapping them in a cycle of increasing commitment.

3.6 Subscription Models

Video game subscription models which are frequently combined with special benefits like priority server access or in-game incentives demand regular payments in order to access game content. These models encourage sustained player engagement while providing developers with recurring money income. For example, in the game Final Fantasy 14 Online (FFXIV) offers two subscription tiers: Standard ($14.99 per month for up to 40 characters) and Entry ($12.99 per month for 8 characters), emphasizing flexibility and value for what each player needs (Lee, 2021).

The success of these models rests on balancing perceived value with psychological incentives. For instance, *FFXIV*'s free trial allows players to explore the base game and first expansion, reducing upfront barriers before transitioning to paid subscriptions (Lee, 2021). It also makes it important for the game developers to keep their players interested for a long time. While improving graphics, storytelling, and gameplay variety can help with that, another approach is to carefully control how fast players advance through the game and design rewards that shape their habits without them even realizing it (Klemm & Pieters, 2017).

3.6.1 Psychology Behind Subscription Models

Similarly to the sunk cost fallacy, players continue subscriptions due to investments of time, money or emotional attachment to in-game progress (Coyle, 2020). They also feel that they are obligated to play the game since they have already bought the subscription, especially if the payment is made annually or quarterly. This also can work together with loss aversion, where players do not want to lose all the items and friends they made while playing that game (Coyle, 2020).

Regular content updates (e.g. *FFXIV*'s quarterly patches) integrate the game into daily routines, making subscriptions feel indispensable. These updates also incite FOMO into players since they may not want to miss playing these updates and might make them not cancel their subscription.

4 Empirical Research

In this section, the author explores the results of a quantitative research carried out via an online questionnaire. The survey builds on theoretical part, specifically player engagement drivers, monetization tactics related to psychological methods and players' perceptions of these tactics.

4.1 Research Objective

The aim of this research is to provide insight into players' perspective about the psychological tactics presented in this thesis, meaning to identify which engagement drivers most strongly influence players' weekly playtime and to assess the impact of monetization tactics on actual player spending. In addition, the study examines how players' awareness of these tactics relates to their perceived enjoyment and demands for transparency.

4.2 Methodology

This survey was conducted via a questionnaire made by the author using Google Forms, which ran from 25.4.2025 until 30.4.2025 and the responses were purely anonymous. The questions in the survey, apart from the demographic-related ones, were all Likert scales. The questionnaire was targeted at video game players of any age or gender and was posted on Facebook, Reddit and Discord. The data collected can be found in appendix A in the form of an Excel sheet.

4.2.1 Research Questions

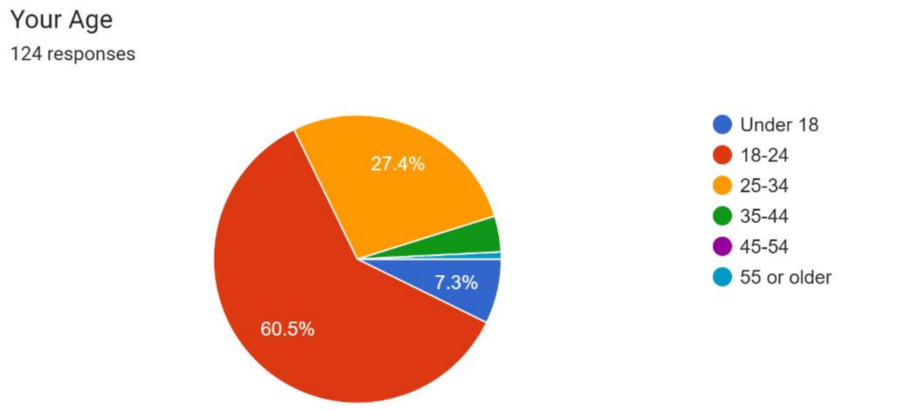
The research questions were stated as this:

- *How do engagement drivers (flow, DDA, limited-time events and reward-seeking motivations) affect players' playtime?*
- *How do monetization mechanics that employ psychological methods affect players' spending habits?*
- *How do players perceive monetization and psychological tactics, and what is their opinion on greater transparency related to these tactics?*

4.3 Results of Quantitative Research

In this quantitative research participated 124 respondents. The first part of the questionnaire was regarding the demographic. The survey was completed by 92 men, 31 women and 1 person identifying as "other."

Figure 15 - *Age of the respondents (Source: Author)*



The most prevalent age groups in this research were 18-24 (60.5%) followed by 25-34 (27.4%), under 18 (7.3%), 35-44 (4%) and lastly 55 or older (0.8%).

4.3.1 How do engagement drivers (flow, DDA, limited-time events and reward-seeking motivations) affect players' playtime?

The second part of the survey focuses on players' weekly playtime and in-game engagement experiences. The questions were related to the flow, DDA, FOMO and operant conditioning mentioned in chapters 1 and 2. The respondents were asked to answer questions (shown in each figure) on a linear 1 to 5 scale where 1 meant "strongly disagree" and 5 meant "strongly agree."

Figure 16 - *Average weekly playtime (Source: Author)*

The distribution of weekly gaming hours reveals that the largest group of participants (35.5%) reported playing for 10-20 hours, followed closely by those gaming 0-10 hours

(28.2%) and 20-40 hours (27.4%) while the smallest group (8.9%) fell into the "more than 40 hours" range. The data is balanced so there was no need to group some of the categories.

Figure 17 – *Perception of time distortion while gaming (Source: Author)*

Group	Average	SD
0–10 hours	3.26	0.95
10–20 hours	3.64	0.97
20–40 hours	3.59	1.05
More than 40 hours	4.36	0.92

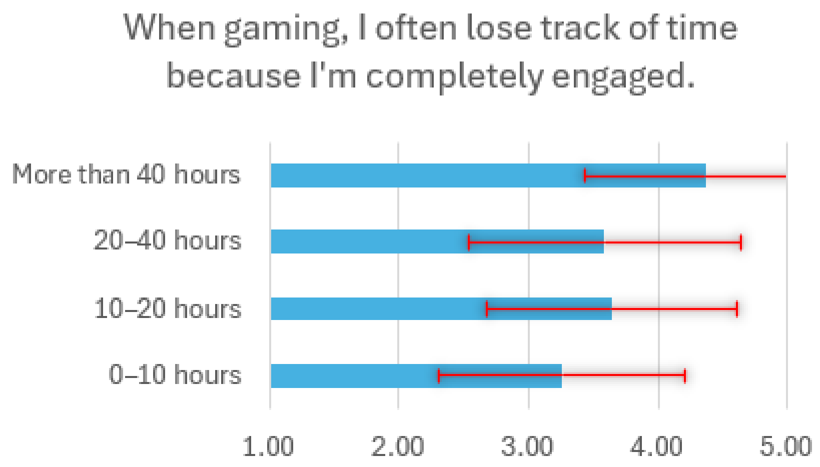
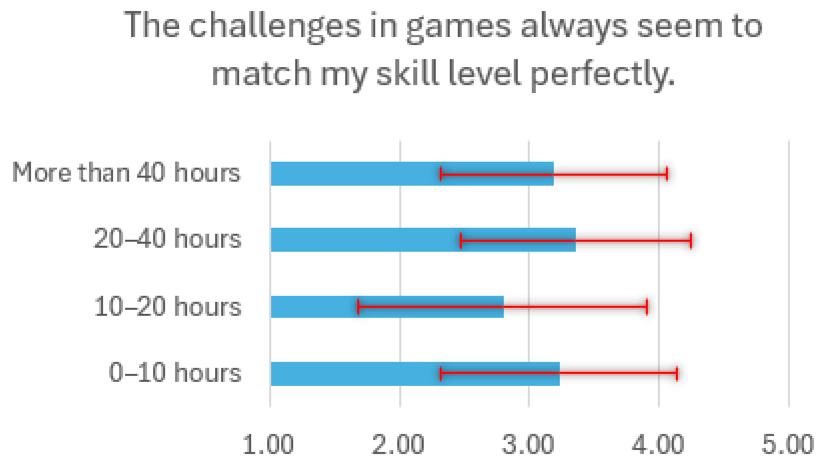


Figure 17 shows (blue columns represent average, red lines represent standard deviation) that players gaming more than 40 hours per week strongly agree they lose track of time (average = 4.36), while casual players (0-10 hours) agree less (average = 3.26). This suggests immersion intensifies with playtime, possibly due to habitual engagement or game design tactics (e.g., flow). However, the small dip in the "20-40 hours" group (average = 3.59) may reflect players balancing gaming with responsibilities, reducing perceived immersion.

While most groups show moderate agreement about losing track of time (standard deviation (SD) less than 1), the 20-40h group's high variability (SD = 1.05) implies that engagement drivers like challenge or rewards may not uniformly affect players in this category. This could reflect external factors (e.g., time constraints) moderating immersion.

Figure 18 – *Perception of challenge-skill balance (Source: Author)*

Group	Average	SD
0–10 hours	3.23	0.91
10–20 hours	2.80	1.11
20–40 hours	3.35	0.88
More than 40 hours	3.18	0.87

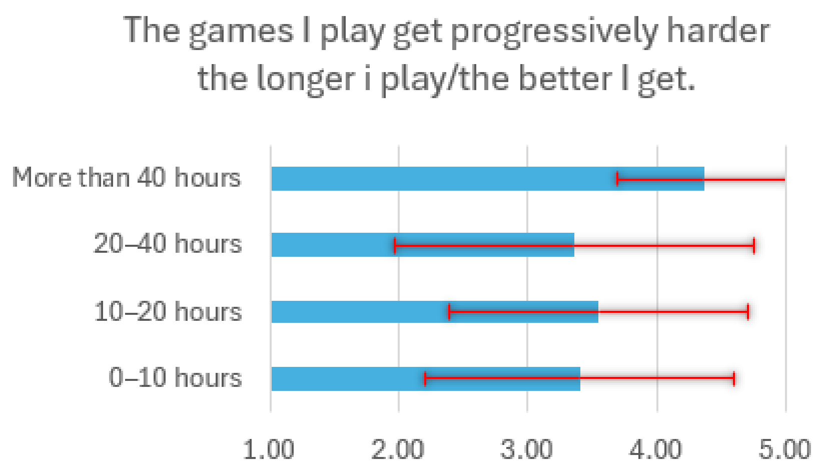


The data reveals mixed perceptions among players regarding how well game challenges align with their skill levels, with high variability across playtime groups. Players in the 20-40 hours per week category report the strongest agreement (average = 3.35) that challenges match their skills, suggesting they benefit from balanced difficulty scaling or adaptive game mechanics. However, this consensus weakens slightly for the more than 40 hours per week group (average = 3.18) potentially due to burnout, repetitive content or heightened expectations from extensive playtime.

The 10-20 hours per week group shows the lowest agreement (average = 2.80) and the highest variability (SD = 1.11), indicating polarized experiences. Some players in this category may struggle with mismatched difficulty spikes, while others might engage with games that align better with their skill level. This divergence could stem from genre diversity (e.g., competitive vs. casual games), inconsistent game design or external factors like time constraints worsening their engagement.

Figure 19 – *Perception of progressive difficulty scaling (Source: Author)*

Group	Average	SD
0–10 hours	3.40	1.19
10–20 hours	3.55	1.15
20–40 hours	3.35	1.39
More than 40 hours	4.36	0.67

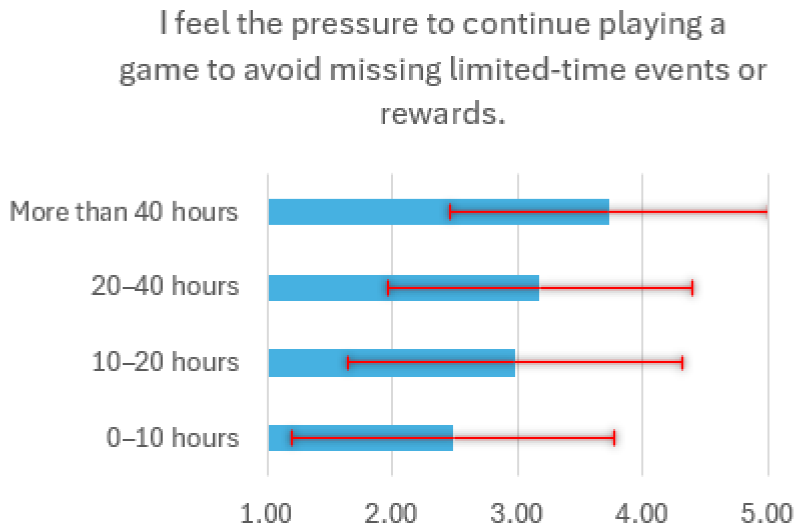


Players gaming more than 40 hours per week demonstrate the strongest agreement (average = 4.36) and the lowest variability (SD = 0.67), indicating near-unanimous satisfaction with how games scale difficulty. This likely reflects their engagement with titles that prioritize dynamic challenge systems, such as competitive multiplayer games or progression-heavy genres requiring sustained skill development.

In contrast, the other groups report moderate agreement alongside high variability. This polarization suggests divergent experiences, some players encounter well-paced difficulty curves, while others face stagnation or poorly balanced challenges, potentially due to repetitive mechanics, skill plateaus or inconsistent game design.

Figure 20 – *Perception of pressure from limited-time events or rewards (Source: Author)*

Group	Average	SD
0–10 hours	2.49	1.29
10–20 hours	2.98	1.34
20–40 hours	3.18	1.22
More than 40 hours	3.73	1.27



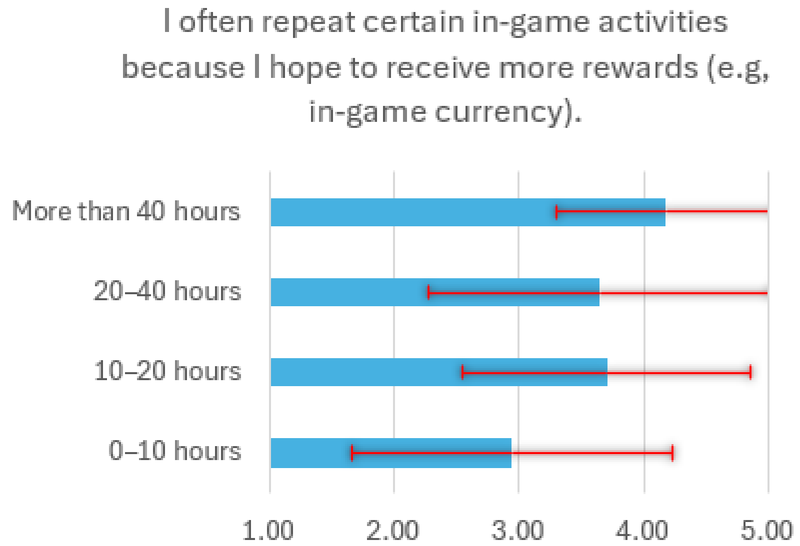
Players gaming more than 40 hours per week report the strongest agreement (average = 3.73) with feeling pressured by limited-time events or rewards, though exhibited variability is high (SD = 1.27). This suggests that while many heavy players experience significant pressure, others may resist or disengage from such mechanics despite their extensive playtime.

Casual players (0-10 hours per week) report the lowest agreement (average = 2.49) but maintain high variability (SD = 1.29), underscoring mixed sentiments, meaning some may ignore events entirely while others may engage in these events.

The upward trend may signal that increased investment in a game amplifies sensitivity to exclusive rewards or time-limited content. However, the persistent variability across all groups suggests that individual factors, such as personal priorities, time constraints or game genre preferences, moderate these pressures. For example, players in competitive or live-service games may feel stronger pressure than those in single-player or casual titles.

Figure 21 - *Engagement with reward-driven repetitive activities (Source: Author)*

Group	Average	SD
0–10 hours	2.94	1.28
10–20 hours	3.70	1.15
20–40 hours	3.65	1.37
More than 40 hours	4.18	0.87



Players gaming more than 40 hours per week report the highest agreement (mean = 4.18) and the lowest variability (SD = 0.87), indicating near-universal engagement with repetitive reward-seeking behaviors. This aligns with their likely participation in games requiring grinding (e.g., MMORPGs, live-service titles).

Casual players (0-10 hours per week) exhibit the weakest agreement (mean = 2.94) and high variability (SD = 1.28), reflecting sporadic or selective engagement with reward systems. This group may prioritize casual enjoyment over repetitive tasks or lack exposure to games demanding extensive grinding.

The escalating agreement with playtime might underscore the role of reward loops in sustaining long-term engagement. The variability in mid-tier groups highlights divergent motivations, meaning some players derive satisfaction from incremental rewards, while others perceive grinding as a chore. For casual players, low agreement suggests limited tolerance for repetition, though the high SD indicates exceptions (e.g., casual games with compelling short-term rewards).

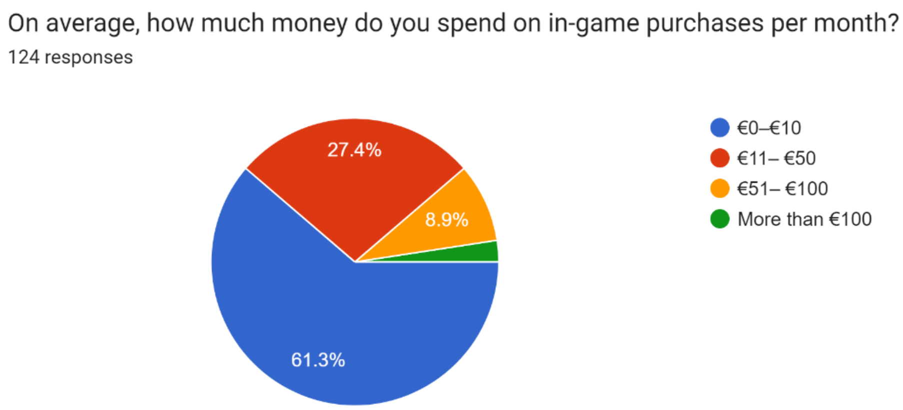
Summary

Players that play for longer periods of time often lose track of time which may signal successful use of game mechanics that leverage the principles of the flow theory. However, respondents only moderately agreed that the in-game challenges matched their skill level which could mean that developers need to change how they introduce the player to the game via tutorials. When it comes to DDA, players with high playtime agreed that games get progressively harder the longer they play/the better they get. This could mean that game makers are implementing DDA in their games successfully. FOMO components in-game are also functioning moderately well since players that play longer are more affected by them. Reward systems (e.g., repetitive tasks) strongly hook heavily invested players but bore casual ones, who may find them tedious.

4.3.2 How do monetization mechanics that employ psychological methods affect players' spending habits?

The next part examined players' monthly expenditure and experiences with monetization in video games. The respondents were asked to answer questions the same way as before, meaning on a linear 1 to 5 scale where 1 meant "strongly disagree" and 5 meant "strongly agree."

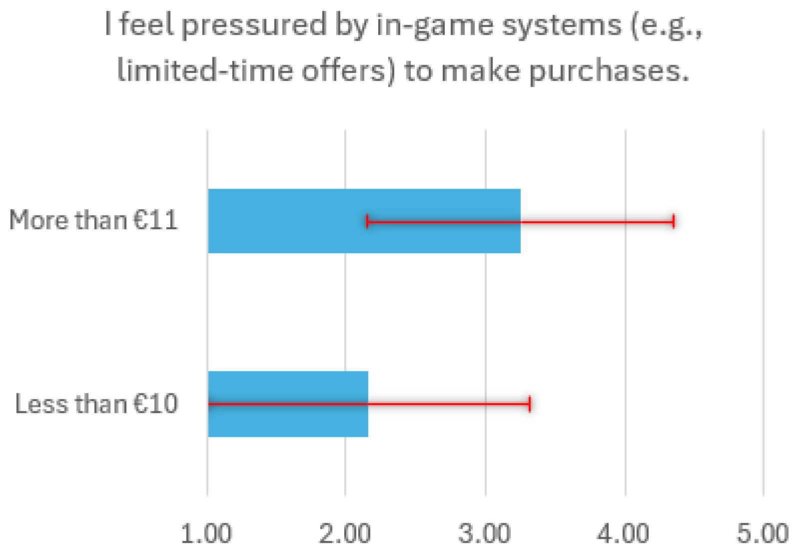
Figure 22 - *Average monthly player expenditure (Source: Author)*



The biggest group of respondents were in the "€0-€10" with more than half of the players falling under that category (61.3%). Because of the categories being disproportionate, the author decided to merge the remaining groups for better clarity and interpretation of the data, meaning creating a spending group called "more than €11."

Figure 23 - *Perceptions on in-game purchase pressure (Source: Author)*

Group2	Average	SD
Less than €10	2.16	1.16
More than €11	3.25	1.10



Players who spend more than €11 monthly report stronger agreement (average = 3.25) with feeling pressured by in-game systems, compared to those spending less than €10 (average = 2.16). The moderate variability in both groups (SD ≈ 1.10-1.16) suggests consistent but not universal experiences.

The disparity between groups may suggest that monetization strategies prioritize high-value players. Frequent spenders (more than €11) likely engage with games employing aggressive tactics (e.g., battle passes), fostering a cycle where spending reinforces sensitivity to pressure. Players who spend less than €10 may avoid or ignore such systems, reflecting either resistance to monetization or preference for non-transactional gameplay.

Figure 24 - *Perceptions on Post-Purchase Obligation (Source: Author)*

Group2	Average	SD
Less than €10	3.47	1.40
More than €11	4.58	0.77

Players who spend more than €11 monthly report near-unanimous agreement (average = 4.58) with feeling obligated to continue playing after a purchase, coupled with low variability (SD = 0.77). This suggests a strong consensus among higher spenders that financial investment compels sustained engagement, likely driven by sunk-cost mentality or rewards tied to recurring purchases (e.g., battle passes).

In contrast, the "less than €10" group shows moderate agreement (average = 3.47) but high variability (SD = 1.40), indicating polarized attitudes. Some casual spenders may feel occasional obligation (e.g., after rare purchases) while others disregard pressure entirely, prioritizing casual play over financial commitments.

Summary

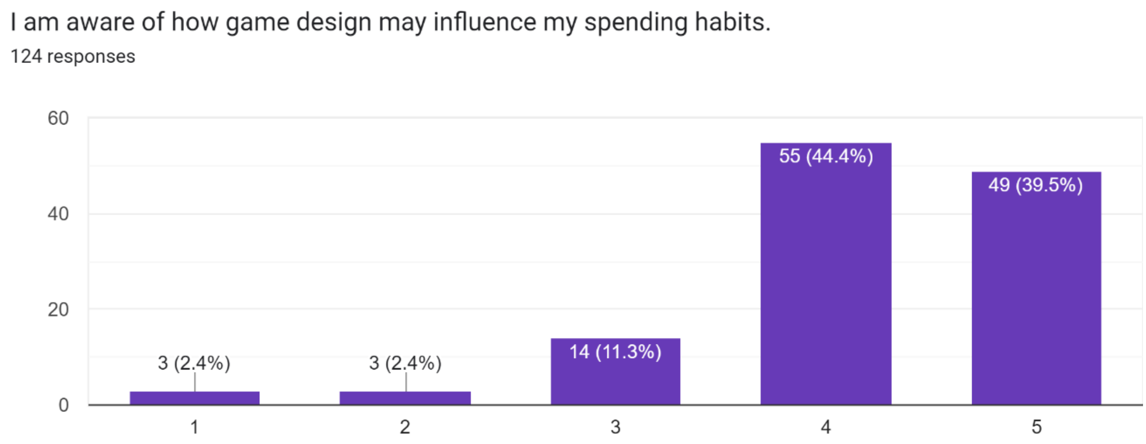
Psychological monetization strategies have been demonstrated to influence spending patterns, although their effects divide players according to their current spending levels. Due to sunk cost thinking, players who spend more money report feeling more pressure to acquire in-game products and almost universally feeling obligated to continue playing after making a purchase.

Because they feel little compulsion to acquire and exhibit inconsistent opinions about post-purchase commitments, players with lower monthly expenditure resist these tactics. Their responses suggest that although most people avoid spending traps, some feel a little guilty or obligated after making a purchase.

4.3.3 How do players perceive monetization and psychological tactics, and what is their opinion on greater transparency related to these tactics?

The last part of the empirical research explores player awareness of in-game monetization tactics, if these tactics diminish their enjoyment of a game and their opinions on transparency regarding these tactics.

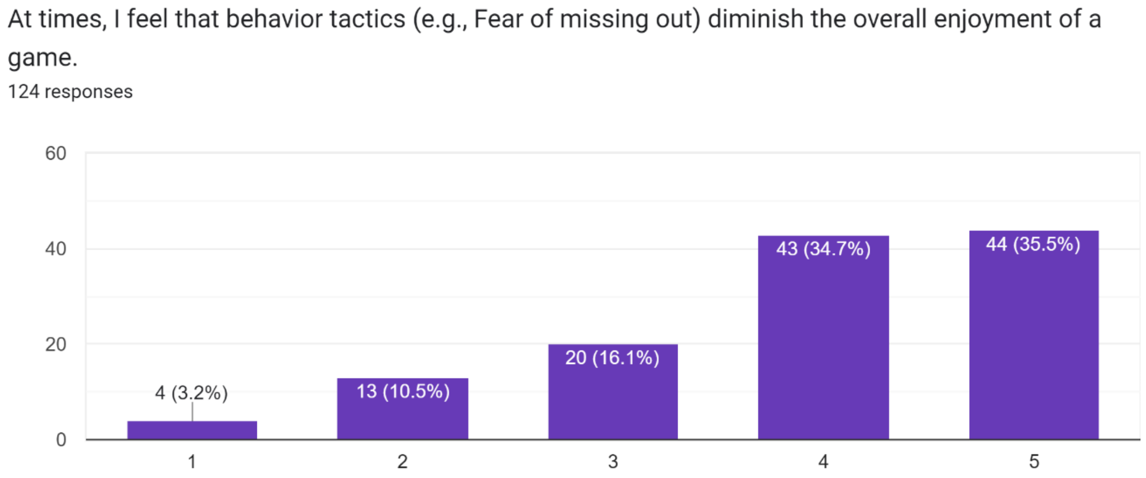
Figure 25 – *Player awareness of game design tactics (Source: Author)*



The data reveals that players exhibit strong awareness of how game design influences their spending habits, with an average agreement score of 4.16. This high average indicates that most respondents acknowledge the role of monetization tactics like limited-time offers, loot boxes, or reward systems in shaping their behavior. The moderate standard deviation (SD = 0.9) suggests that while responses cluster around the upper end of the scale, there is still notable variability in how players perceive these influences.

The average score of 4.16 leans close to "strongly agree," reflecting widespread recognition of manipulative design elements. For instance, players likely understand how mechanics like daily login rewards, battle passes or countdown timers nudge spending. However, the SD of 0.9 highlights that a subset of players, particularly those with lower scores (e.g., ratings of 1-3), either downplay these tactics or remain ambivalent.

Figure 26 – *Player perception of how behavior tactics affect their enjoyment (Source: Author)*



The average score of 3.89 signals that most players associate tactics like limited-time events or exclusive rewards with reduced enjoyment. Over 70% of respondents are in agreement and view these mechanics as disruptive. In contrast, nearly 30% express tolerance, possibly reflecting varied perspectives influenced by factors such as game genre (e.g., live-service vs. narrative-driven titles) or personal resilience to design pressures. The standard deviation of 1.11 highlights this divergence.

Figure 27 - *Player perspectives on greater monetization transparency in video games (Source: Author)*

Players exhibit moderate uncertainty (average = 3.37) toward whether transparency about in-game monetization would influence their decisions to play or spend, with high variability (SD = 1.3) emphasizing diverse opinions. While 48.4% agree that transparency matters, 28.3% dismiss its impact and 23.4% remain neutral. Transparency alone may not curb spending habits driven by impulse, social pressure or FOMO.

Summary

Respondents are broadly aware of monetization tactics and acknowledge their role in influencing spending habits, with most agreeing these designs intentionally shape behavior. Players are skeptical when asked about transparency regarding monetization strategies. Approximately half of the respondents think openness would influence their choices to play or spend, whereas a sizable percentage downplay its significance and around a quarter of the players are neutral or uncertain. This suggests that although honesty is valued by certain players, spending patterns influenced by impulse, sunk cost fallacy or FOMO may not be changed by mere transparency.

5 Conclusions

The bachelor thesis set out a goal to identify the application of behavioral psychology in video game design to prolong player engagement and drive in-game spending, synthesizing fragmented research into a unified overview and offered empirical insights into player experiences. These goals were accomplished, and the paper could help players, developers and regulators gain valuable understanding about these behavioral tactics.

The theoretical section described findings on how gaming mechanics leverage psychological principles to maintain player engagement. The paper first described the core principles of flow theory and explained how video game mechanics, such as “Dynamic Difficulty Adjustment” (DDA) and immersive audio-visual elements, are used to sustain optimal engagement states. The second chapter focused on the psychological methods with exploitative nature, such as FOMO, operant conditioning, the sunk cost fallacy and the endowment effect. Lastly, the theoretical part analyzed various types of predatory monetization systems and how they incorporate the aforementioned exploitative methods, providing concrete examples and outlining relevant regulatory actions regarding them.

The quantitative research shed light on players' experiences with these strategies, their knowledge of them and their opinions about increased transparency in relation to these strategies. Monthly spending and weekly playtime were used to group the participants. Despite generally acknowledging manipulative design, key findings indicate that players who spent more money and played for longer periods of time reported being more vulnerable to pressure-driven in-game mechanisms including FOMO and sunk cost commitments. Additionally, players were found to be completely immersed which could be related to flow and also the respondents agreed that games get harder the longer the play or the better they get which could mean that developers are utilizing DDA. Although there was some demand for more monetization transparency, several respondent had doubts about its real-world implications.

Nevertheless, the research had its limitations. The quantitative study relied on self-reported data which may skew the actual effect of the psychological methods. Emotional effects like regret or compulsion could be investigated in more detail using qualitative interviews and working with developers could test ethical frameworks like spending limitations or opt-in pressure mechanisms. Fast innovation in game design means that emerging predatory monetization models may not yet be fully captured here. Moreover, regional differences in regulation and cultural attitudes toward gaming suggest that the ethical landscape could look quite different across the world.

In order to improve gaming in an ethical manner and reduce exploitative monetization methods like battle passes and gacha mechanics, developers should give priority on employing psychological strategies like DDA. Regarding this issue, regulators should try to promote international standards. Harmonized international laws combined with cultural flexibility can prevent exploitation and promote innovation, guaranteeing that player welfare keeps pace with industry growth.

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